Finite cores stabilise Thomson's heptagon

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Abstract

Seven identical point vortices at the vertices of a regular heptagon form Thomson's heptagon. It sits exactly on the boundary between stable and unstable rings: its linearisation has a degenerate (zero) critical mode. We ask which way this degeneracy breaks when the point vortices are replaced by small uniform vortex patches of radius r on a ring of radius R. We show by formal asymptotics that the correction to the ring's linear stability has no $(r/R)^2$ term. It enters at order $(r/R)^4$ through the elliptical deformation that each patch undergoes in the strain of the others. We derive the order-$(r/R)^4$ Fourier blocks of the ring for every N and every Bloch mode m as explicit polynomials in (N, m). For the heptagon the critical stiffness becomes $a_3^{\perp} = +14\,(r/R)^4$, so its sign is positive and small patches make the heptagon *linearly stable*. Starting from the lattice-sum form of the blocks, the algebra is checked in Lean 4: the closed forms of the lattice sums over roots of unity, the block polynomials for all N, the values $(14, 56, 16)$ at $N = 7$, and the reduction of stability to the sign of $a_3^{\perp}$. The reduction of the effective Hamiltonian to that lattice-sum form is done by hand and cross-checked by computer algebra. The effective Hamiltonian itself rests on formal asymptotics and is not proved. A contour-dynamics code with an exact (complex-step) Jacobian and a Bloch decomposition confirms the block coefficients for $N = 4, \ldots, 8$ to about four significant figures. It also finds the steady heptagon of patches linearly stable for $0.02 \le r/R < 0.26138$ (numerically), and as $r \to 0$ by the r^4 asymptotics (formally); at 0.26138 an elliptical shape mode collides with a displacement mode and a complex quartet appears. The steady branch ends near $r/R = 0.2802$, apparently at a fold. Our conclusion contradicts the abstract of Dhanak (1992), whose small-core asymptotics give instability for $N \ge 7$, and a secondary report of Dritschel (1985); we could not obtain Dhanak's full text and cannot locate the discrepancy. A short section gives the corresponding $(r/R)^4$ coefficients on the sphere and the hyperbolic plane.

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1 Introduction

The problem. N identical point vortices at the vertices of a regular N-gon rotate rigidly about its centre. Thomson [20] studied the stability of this configuration. In the form corrected by Havelock [7], the result is that the ring is linearly stable for $N \leq 6$, linearly unstable for $N \geq 8$, and neutral for $N = 7$: the critical Fourier mode $m = 3$ of the heptagon has a vanishing stiffness, so the linearisation has a double zero eigenvalue on that mode (Proposition 2.2). The heptagon is in fact Lyapunov stable as a point-vortex system (Cabral and Schmidt [2], Kurakin and Yudovich [11]), but that stability is decided by quartic terms. Any perturbation of the model that enters at quadratic order decides its *linear* fate.

Point vortices idealise vortices with small cores. The most natural core model is the uniform vortex patch: a disc-shaped region of constant vorticity, evolving under the two-dimensional Euler equations. Does a ring of seven small patches inherit the stability of the point heptagon, or does the finite core size tip it to instability? Steady rings of patches near any Thomson polygon exist (García [5]; see also [8, 6]), so the question is well posed.

What was known. Dritschel [4] computed corotating uniform vortices for $N = 2, \ldots, 8$ and their linear stability. Dhanak [3] studied the small-core limit analytically; his abstract states that the system "is stable if $N < 7$ and unstable if $N \geq 7$", in general agreement with Dritschel except "in a region of parameter space where a high degree of accuracy is required". Polvani and Dritschel [16, p. 52] write that "when $N = 7$, any finite size leads to instability – see Dhanak 1992". Llewellyn Smith and Crowdy [12] report that in Dritschel's study the heptagon "becomes unstable to a displacement instability when the point vortices are replaced by vortex patches, even very small ones".

What we find. The opposite (Section 7 discusses the conflict).

- For a ring of N patches the finite-core correction to every Fourier block of the linearisation has no $(r/R)^2$ term. The first correction is of order $(r/R)^4$, and we give it in closed form for all N and all Bloch modes m (Theorem 4.3). For the heptagon's critical mode,

$$a_3^{\perp} = 14\,(r/R)^4, \qquad a_3^{\parallel} = 6 + 56\,(r/R)^4, \qquad \varpi_3 = 16\,(r/R)^4$$

(Corollary 4.5). Since $a_3^{\perp} a_3^{\parallel} > 0$, the slow eigenvalues are purely imaginary: small patches make the heptagon linearly stable. These coefficients are derived from an effective Hamiltonian obtained by formal asymptotics (status **Derived**). Everything after that Hamiltonian is exact algebra, proved by hand; from the lattice-sum form on it is also checked in Lean, and the reduction to that form is cross-checked by computer algebra.

- A contour-dynamics computation reproduces the order-$(r/R)^4$ coefficients of all blocks for $N = 4, \dots, 8$ (Table 1). It also finds the steady heptagon of patches linearly stable for $0.02 \le r/R < 0.26138$ (**Numerical**); for $r \to 0$ stability rests on the r^4 asymptotics (**Derived**). The instability at 0.26138 is a collision between a displacement mode and an elliptical shape mode (Section 5). An independent referee code and a nonlinear simulation agree (Section 5.4).

Status labels. Every claim carries one of the labels **Proved** (hand proof here), **Lean** (checked in Lean 4 with Mathlib [19]; module and theorem named), **Certified** (exact computer arithmetic; script named), **Derived** (formal asymptotics), **Numerical** (floating point). The file `README.md` in the paper's directory maps every numbered statement to its proof, Lean theorem, script and output.

2 The point-vortex ring and its Fourier blocks

Units. We use the normalisation in which a point vortex of circulation Γ has physical circulation $2\pi\Gamma$. The Hamiltonian of N point vortices $z_k \in \mathbb{C}$ is then

$$H = -\sum_{j<k} \Gamma_j \Gamma_k \log |z_j - z_k|,$$

with symplectic form $\sum_k \Gamma_k \, dx_k \wedge dy_k$, so that $\Gamma_k \dot{x}_k = \partial H/\partial y_k$, $\Gamma_k \dot{y}_k = -\partial H/\partial x_k$. All patches below have $\Gamma = 1$. A uniform patch of radius r then has vorticity $2/r^2$. Time scales as R^2 and only r/R matters, so we set the ring radius $R = 1$ and write

$$\epsilon := (r/R)^4.$$

Rotating frame and log-polar coordinates. Let $J = \frac{1}{2}\sum_k |z_k|^2$ and $F = H - \Omega J$. A ring rotating rigidly with angular velocity $-\Omega$ is a critical point of F. With $\zeta = e^{2\pi i/N}$ we write

$$z_k = \zeta^k e^{s_k + i t_k}, \qquad k = 0, \dots, N-1,$$

so s_k is a radial and t_k an angular displacement. In these coordinates the symplectic form at the ring is $\sum_k ds_k \wedge dt_k$, and the linearised flow is $\frac{d}{dt}(s,t) = \mathsf{J}\nabla^2 F\,(s,t)$ with $\mathsf{J} = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ acting on each pair (s_k, t_k).

Bloch blocks. The Hessian $\nabla^2 F$ at the ring is invariant under the cyclic shift $k \mapsto k+1$. On the Bloch mode $(s_k, t_k) = (\hat{s}, \hat{t})\,\zeta^{mk}$ it acts as the 2×2 Hermitian matrix

$$B_m = \sum_{j=0}^{N-1} \nabla^2_{0,j} F \; \zeta^{mj} = \begin{pmatrix} a_m^{\perp} & i\varpi_m \\ -i\varpi_m & a_m^{\parallel} \end{pmatrix}, \tag{1}$$

where $\nabla^2_{0,j} F$ is the 2×2 block of second derivatives with respect to (s_0, t_0) and (s_j, t_j). The form in (1) (real diagonal, purely imaginary off-diagonal) follows from the reflection symmetry $k \mapsto -k$, $t \mapsto -t$ of the ring. The linearised flow decouples into the 2×2 systems $\frac{d}{dt}(\hat{s}, \hat{t}) = \mathsf{J} B_m (\hat{s}, \hat{t})$.

Lemma 2.1 (Spectrum of a block)**.** $\det(\lambda - \mathrm{J}B_m) = (\lambda + i\varpi_m)^2 + a_m^\perp a_m^\parallel$. *Hence the eigenvalues on the block are* $\lambda = -i\varpi_m \pm \sqrt{-a_m^\perp a_m^\parallel}$. *They are purely imaginary if* $a_m^\perp a_m^\parallel > 0$, *and one has positive real part if* $a_m^\perp a_m^\parallel < 0$.

[Proved; Lean] The determinant is a 2×2 computation. Lean: `Heptagon.BlockSpectrum`, theorems `charpoly_block`, `block_eigs_imaginary`, `block_unstable`, `block_eigs_explicit`.

Note that ϖ_m only shifts the pair along the imaginary axis. For every $m \neq 0$ the stability of the block is decided by the sign of $a_m^\perp a_m^\parallel$, and $a_m^\parallel > 0$ in all cases below, so by the sign of $a_m^\perp$.

Proposition 2.2 (Point-vortex blocks)**.** *For the point-vortex ring,* $\Omega = -\frac{N-1}{2}$, $\varpi_m = 0$, *and for* $1 \le m \le N-1$

$$a_m^\perp = N - 1 - \tfrac{1}{2}m(N-m), \qquad a_m^\parallel = \tfrac{1}{2}m(N-m).$$

The critical mode $m = \lfloor N/2 \rfloor$ *has* $a^\perp = N - 1 - \frac{1}{2}\lfloor N^2/4 \rfloor$, *which is positive for* $N \le 6$, *zero for* $N = 7$ *and negative for* $N \ge 8$. *For* $N = 7$, $m = 3$*:* $(a_3^\perp, a_3^\parallel) = (0, 6)$.

[Proved; Lean] Proof in Section 4 (the case $\epsilon = 0$ of Proposition 4.2). Lean: `Heptagon.RingBlocks.pv_blocks` (from the lattice sums) and `heptagon_blocks`. This is the classical Thomson–Havelock result in block form.

3 Finite cores: the effective Hamiltonian

We now replace each point vortex by a uniform patch of circulation 1 and radius r. The dynamics is the Euler flow of the patches, an infinite-dimensional Hamiltonian system on an isovortical leaf. Its energy, in these units, is $H[q] = -\frac{1}{2}\iint \log|x-y|\, q(x)q(y)\, dx\, dy$, where q is the vorticity divided by 2π. Each patch has two kinds of degrees of freedom: the position of its centroid, and the shape of its boundary. We write the boundary of patch k in polar form about its centroid c_k,

$$\rho_k(\theta) = r\big(1 + \mathrm{Re}\sum_{n\ge 2} \varepsilon_{k,n} e^{-in\theta}\big) + O(\varepsilon^2),$$

with complex shape amplitudes $\varepsilon_{k,n}$. The $O(\varepsilon^2)$ term restores the area.

3.1 Rigid discs are point vortices

Lemma 3.1 (Newton's theorem for the logarithm)**.** *If* $0 < \varrho < |c-a|$, *the average of* $\log|z-a|$ *over the circle* $|z-c| = \varrho$ *equals* $\log|c-a|$. *If two circles* $|z-c_1| = \varrho_1$, $|w-c_2| = \varrho_2$ *satisfy* $\varrho_1 + \varrho_2 < |c_1 - c_2|$, *the double average of* $\log|z-w|$ *equals* $\log|c_1 - c_2|$.

[Lean] `Heptagon.Newton`: `circleAverage_log_far`, `circleAverage_log_two_circles`. The proof uses Mathlib's circle-average formula for $\log|z-a|$.

Averaging over the radii $0 < \varrho \le r$ (an elementary step done by hand) gives the following. The stream function of a circular patch outside the patch is that of a point vortex at its centre. The mutual energy of two disjoint circular patches equals that of two point vortices. The self-energy of a circular patch is a constant. So if every patch is constrained to remain a disc, the dynamics of the centres is exactly the point-vortex dynamics, *to all orders in* r. Finite-core effects therefore enter only through the shape modes.

3.2 The strain-induced ellipse

Lemma 3.2 (Energy of a weakly deformed patch)**.** *Let a patch of circulation Γ and radius r sit in an external stream function P whose complex trace-free Hessian at the centroid is $M = \frac{1}{2}(P_{xx} - P_{yy}) - iP_{xy}$. To second order in the amplitude of its elliptical mode $n = 2$, the energy of the patch is*

$$E(\varepsilon) = E_0 - \frac{\Gamma^2}{4}|\varepsilon|^2 + \frac{\Gamma r^2}{2}\,\mathrm{Re}(M\varepsilon) + O(r^4|\varepsilon|\,|\nabla^4 P|)\,.$$

More generally, mode n has self-energy $-\Gamma^2\frac{n-1}{2n}|\varepsilon_n|^2$ and couples linearly, with a factor r^n, to the n-th Taylor coefficient of P.

[Proved] The self-energy of the ellipse with semi-axes $re^{\pm s}$ is $E_0 - \frac{\Gamma^2}{2}\log\cosh s$ (Kirchhoff, Love [13]; see [17]). Its boundary is $r(1 + s\cos 2\theta) + O(s^2)$, which gives $-\frac{\Gamma^2}{4}s^2$. For the coupling, the patch has density $\Gamma/(\pi r^2)$ and its interaction energy is $\int q\,P$. In polar coordinates about the centroid, the trace-free quadratic part of P is $\frac{1}{2}\varrho^2\,\mathrm{Re}(Me^{2i\theta})$; the trace vanishes because P is harmonic. The added region has thickness $r\,\mathrm{Re}(\varepsilon e^{-2i\theta})$. So

$$\delta E = \frac{\Gamma}{\pi r^2}\int_0^{2\pi} \tfrac{1}{2}r^2\,\mathrm{Re}(Me^{2i\theta})\,r^2\,\mathrm{Re}(\varepsilon e^{-2i\theta})\,d\theta = \frac{\Gamma r^2}{2}\,\mathrm{Re}(M\varepsilon).$$

The area correction at second order multiplies the mean value of P over the circle, which is $P(c)$ by Lemma 3.1, and cancels against the second-order area of the $n = 2$ term. □

Proposition 3.3 (Effective Hamiltonian)**.** *For N patches of circulation 1 and radius r with centres z_k, the slow (centre) dynamics is, to order r^4, the point-vortex dynamics with the Hamiltonian*

$$H_{\mathrm{eff}} = H + \frac{r^4}{4}\sum_k |M_k|^2 + O(r^6), \qquad M_k = \sum_{l\neq k}\frac{1}{(z_k - z_l)^2}, \tag{2}$$

and the point-vortex symplectic form. Each patch is an ellipse with complex amplitude $\varepsilon_k = r^2\overline{M_k}$, and aspect ratio $1 + 2r^2|M_k| + O(r^4)$ (the small-strain Moore–Saffman law [15, 9]). There is no r^2 term.

[Derived; algebra Lean] The argument has three steps. (i) The shape modes are fast. Mode n of an isolated patch has frequency $(n-1)/r^2$, while the centre motion has frequencies of order one. (ii) The symplectic (Kirchhoff–Kostant–Souriau) form on the isovortical leaf is block-diagonal between centre translations and the centrally symmetric shape modes. This holds because every momentum function of a rigid motion has Fourier content $|n| \le 1$ about a patch centre. The coupling between centres and shapes is therefore only through the energy, and by Lemma 3.2 it is of order r^2. (iii) Eliminate the fast modes adiabatically: put each shape at the stationary point of the energy at fixed centres, which is the Schur complement of the Hessian. By Lemma 3.2 with $\Gamma = 1$ and the external stream function $P_k = -\sum_{l\neq k}\log|z - z_l|$, whose trace-free Hessian is M_k, the stationary amplitude is $\varepsilon = r^2\overline{M_k}$ and the stationary value is $+\frac{r^4}{4}|M_k|^2$, independently of Γ. Modes $n \ge 3$ contribute $O(r^{2n}) = O(r^6)$. Curvature of the boundary, cubic terms, and quadrupole–quadrupole interactions between patches are also $O(r^6)$, and so is the gyroscopic remainder of the elimination (fast frequencies $\sim r^{-2}$, coupling $\sim r^2$). The remainder is order-counted, not bounded; that is why the status is *Derived*. In Lean (`Heptagon.Elimination`) we check the algebra of step (iii): completing the square (`complete_square`, `elim_max`), the values $r^4|M|^2/4$ and $\varepsilon = r^2M/\Gamma$ (`patch_elimination`), the aspect ratio (`aspect_ratio`), and the Schur-complement identity for the quadratic form (`schur_elimination`). The same effective Hamiltonian follows from an elliptical-moment model in the spirit of [14].

Remark 3.4. The asymptotics assume that the steady patch ring exists and is close to the discs. This is proved by García [5, Thm. 1.1] for every $N \geq 2$ and all small patch sizes, with boundaries at least C^1 (the construction is in $C^{1+\alpha}$ spaces; the paper's expansions also use smoothness of the boundary, which we do not verify here beyond C^1). Her ansatz is N-fold symmetric: all patches are rotated copies of one, $D_m = e^{2\pi i m/N} D_0$, and the implicit function theorem is applied to the shape of D_0 together with Ω; the relevant linearised operator is an isomorphism independently of N (we checked this in the arXiv version). The $N = 7$ degeneracy lives in the symmetry-breaking mode $m = 3$, which this symmetric existence problem does not see, so existence is covered for $N = 7$. The contour-dynamics solutions of Section 5 agree.

4 The Fourier blocks of the effective Hamiltonian

4.1 Lattice sums

For $a \geq 0$ and $n \in \mathbb{Z}$ let

$$T_a(n) = \sum_{j=1}^{N-1} \frac{\zeta^{jn}}{(1-\zeta^j)^a}, \qquad \zeta = e^{2\pi i/N}.$$

Lemma 4.1 (Lattice sums)**.** *(a)* $T_a(n+N) = T_a(n)$. *(b)* $T_a(n) - T_a(n+1) = T_{a-1}(n)$ *for* $a \geq 1$. *(c)* $\sum_{n=0}^{N-1} T_a(n) = 0$. *(d)* $T_0(n) = -1$ *for* $0 < n < N$. *(e) For* $1 \leq n \leq N$,

$$\begin{aligned}
T_1(n) &= \tfrac{1}{2}(2n - N - 1),\\
T_2(n) &= -\tfrac{1}{12}\big(N^2 - 6Nn + 6N + 6n^2 - 12n + 5\big),\\
T_3(n) &= \tfrac{1}{24}\big(2N^2 n - 3N^2 - 6Nn^2 + 18Nn - 12N + 4n^3 - 18n^2 + 24n - 9\big),\\
T_4(n) &= \tfrac{1}{720}\big(N^4 - 30N^2n^2 + 120N^2 n - 110N^2 + 60Nn^3 - 360Nn^2 + 660Nn - 360N\\
&\qquad - 30n^4 + 240n^3 - 660n^2 + 720n - 251\big).
\end{aligned}$$

In particular $T_1(0) = \frac{N-1}{2}$, $T_2(0) = -\frac{(N-1)(N-5)}{12}$, $T_3(0) = -\frac{(N-1)(N-3)}{8}$, $T_4(0) = \frac{(N-1)(N^3+N^2-109N+251)}{720}$.

[Proved; Lean] (a)–(d) are one-line computations with $\zeta^N = 1$ and the geometric sum. For (e), write $D(n) = T_a(n) - P_a(n)$, where P_a is the stated polynomial. The polynomials satisfy $P_a(n) - P_a(n+1) = P_{a-1}(n)$ (with $P_0 = -1$) and $\sum_{n=1}^{N} P_a(n) = 0$; the latter is shown by exhibiting an antiderivative Q_a with $Q_a(0) = Q_a(N) = 0$. By (b), D is then constant on $1 \leq n \leq N$, and by (a) and (c) its sum over a period is 0. So $D \equiv 0$, by induction on a. Lean: `Heptagon.LatticeSums`, theorems `T_add_N`, `T_sub_succ`, `sum_T`, `T_zero`, `T_eq_of` (the induction), `T1_closed`–`T4_closed`, `sum_inv_one_sub`, `sum_inv_one_sub_sq`; all for every $N \geq 1$ and every primitive N-th root of unity. The closed forms also agree with direct 50-digit summation for $N \leq 15$, $a \leq 6$ (`code/tsums.py`).

4.2 Reduction of the blocks to lattice sums

Write $p_j = (1-\zeta^j)^{-1}$; then $\sum_j p_j^a \zeta^{jn} = T_a(n)$, and $\zeta^j p_j^2 = p_j^2 - p_j = -1/(4\sin^2(\pi j/N))$.

Proposition 4.2 (Blocks as lattice sums)**.** *For* $F = H_{\text{eff}} - \Omega J$ *at the ring of radius* 1*, with* $\Omega = \Omega_0 + \epsilon\Omega_1 + O(\epsilon^{3/2})$*, and* $1 \leq m \leq N-1$*:*

$$\begin{aligned}
&\Omega_0 = -\tfrac{N-1}{2}, \qquad \Omega_1 = -\mu^2, \qquad \mu := T_2(0),\\
&a_m^{\perp} = N - 1 + g(0) - g(m) + \epsilon\big[\tfrac{1}{4}(\Lambda_+^2 + \Lambda_-^2) + \mu\kappa + 2\mu^2\big],\\
&a_m^{\parallel} = g(m) - g(0) + \epsilon\big[\tfrac{1}{4}(\Lambda_+^2 + \Lambda_-^2) - \mu\kappa\big], \qquad \varpi_m = \epsilon\,\tfrac{1}{4}(\Lambda_+^2 - \Lambda_-^2),
\end{aligned}$$

up to $O(r^6)$, where $g(n) = T_2(n) - T_1(n)$ and

$$\Lambda_\pm = -2\big(T_3(0) - T_3(1 \pm m)\big), \qquad \kappa = -T_2(0) + 3\big(T_4(0) - T_4(1-m) - T_4(1+m) + T_4(2)\big).$$

[Proved (given (2))] The proof is in Appendix A: second-order Taylor expansion of each pair term, then evaluation of the Hermitian form on Bloch modes. As an independent check, the program `code/blocks.py` performs the same reduction mechanically: it does a symbolic Taylor expansion of the local terms, maps monomials to lattice sums, and evaluates them exactly. It reproduces the formulas above for symbolic (N, m) (`plane_general.py`, `blocks.py`), and the exact blocks for every $N = 2, \dots, 12$ and $0 \le m \le N/2$ (`outputs/blocks.out`). A third, independent check (direct 40-digit differentiation of H_{eff}, no lattice sums) is in Section 5.4.

Theorem 4.3 (The r^4 blocks of the planar ring)**.** *For every $N \ge 2$ and $1 \le m \le N-1$, the lattice-sum expressions of Proposition 4.2 equal*

$$\begin{aligned}
a_m^\perp &= N - 1 - \tfrac{1}{2}m(N-m) + \epsilon\, d^\perp(N,m), \qquad a_m^\parallel = \tfrac{1}{2}m(N-m) + \epsilon\, d^\parallel(N,m), \qquad \varpi_m = \epsilon\, d^\varpi(N,m),\\
d^\parallel &= \tfrac{1}{144}\, m^2(N-m)^2(5N^2 - 8Nm - 18N + 8m^2 + 33),\\
d^\varpi &= -\tfrac{1}{36}\, m(N-m)(N-2m)(-N^2 - 3Nm + 6N + 3m^2 - 5),\\
d^\perp &= \tfrac{1}{144}\big(- N^4m^2 + 6N^4 - 6N^3m^3 + 18N^3m^2 + 24N^3m - 72N^3 + 23N^2m^4 - 36N^2m^3\\
&\qquad - 21N^2m^2 - 144N^2m + 276N^2 - 24Nm^5 + 18Nm^4 - 6Nm^3 + 144Nm^2 + 120Nm - 360N\\
&\qquad + 8m^6 + 3m^4 - 120m^2 + 150\big),
\end{aligned}$$

and $\Omega = -\frac{N-1}{2} - \frac{(N-1)^2(N-5)^2}{144}\epsilon$: the ring turns faster by $\frac{(N-1)^2(N-5)^2}{144}(r/R)^4$ (in units R^{-2}), except for $N = 5$, where the strain at each patch vanishes ($\mu = 0$).

[Lean] `Heptagon.RingBlocks.r4_blocks` (all four identities and μ, Ω_1, for every $N \ge 2$ and $1 \le m \le N-1$) and `pv_blocks`. The proof substitutes Lemma 4.1(e), which needs $1 \le m+1 \le N$ and $1 \le N+1-m \le N$, and finishes with `ring`.

Combining Theorem 4.3 with Proposition 3.3 gives the order-$(r/R)^4$ blocks of the patch ring (status **Derived**, because of Proposition 3.3). In particular no block has an $(r/R)^2$ term.

Corollary 4.4 (The critical mode)**.** *At $m = \lfloor N/2 \rfloor$ the critical stiffness is $a^\perp = N - 1 - \frac{1}{2}\lfloor N^2/4 \rfloor + \mathfrak{A}_N\,\epsilon$, with*

$$\mathfrak{A}_N = -\frac{N^6 - 6N^5 - 65N^4 + 576N^3 - 1632N^2 + 1920N - 800}{768} \qquad (N \text{ even}),$$

$$\mathfrak{A}_N = -\frac{(N-1)^2(3N^4 - 12N^3 - 230N^2 + 1316N - 1925)}{2304} \qquad (N \text{ odd}).$$

$\mathfrak{A}_N = \frac{1}{16}, \frac{2}{9}, \frac{11}{8}, 5, \frac{491}{48}, 14, -\frac{45}{8}, -\frac{556}{9}$ for $N = 2, \dots, 9$. Moreover $\mathfrak{A}_N > 0$ for $N \le 7$ and $\mathfrak{A}_N < 0$ for $N \ge 8$. Thus finite cores strengthen the point-vortex verdict for $N \le 6$ and $N \ge 8$, and decide it for $N = 7$.

[Lean] `A_even`, `A_odd`, `A_values`, `A_even_neg`, `A_odd_neg`. For $N \ge 8$ write $N = 8 + t$: the numerators become polynomials in t with positive coefficients. For $N = 2, 3$ the critical mode is $m = 1$, and the values are those of $m = 1$. The closed forms hold for all N given (2).

Corollary 4.5 (The heptagon)**.** *For $N = 7$, $m = 3$: $(a_3^\perp, a_3^\parallel, \varpi_3) = (14\epsilon,\ 6 + 56\epsilon,\ 16\epsilon) + O(r^6)$ and $\Omega = -3 - \epsilon$. For the truncated block with $\epsilon > 0$ both eigenvalues are purely imaginary:*

$$\lambda_\pm = i\big(- 16\epsilon \pm \sqrt{14\epsilon(6 + 56\epsilon)}\big).$$

For the patch heptagon this means $s_3 := \left(\frac{\lambda_+ - \lambda_-}{2}\right)^2 = -a_3^\perp a_3^\parallel = -84\,(r/R)^4 + O(r^6)$ *and* $\frac{i}{2}(\lambda_+ + \lambda_-) = \varpi_3 = 16\,(r/R)^4 + O(r^6)$*. So the slow pair is* $\lambda = \pm i\sqrt{84}\,(r/R)^2 + O(r^4)$*, and it is purely imaginary to the order computed: the heptagon of small patches is linearly stable.*

[Lean for the algebra; Derived for the conclusion] Lean: `heptagon_blocks`, `heptagon_blocks_exp` (with $\zeta = \exp(2\pi i/7)$), `heptagon_truncated_stable`, `heptagon_pv_degenerate`. The other blocks of the heptagon have $a_m^\perp, a_m^\parallel > 0$ at $\epsilon = 0$, so they stay stable for small r. The $m = 0$ block carries the rotation symmetry, and the $m = 1, 6$ blocks carry the translation pair at $\pm i|\Omega|$.

Remark 4.6 (What the order counting does and does not give). The slow pair has size r^2, while the neglected terms in $a_3^\perp$ are $O(r^6)$. These change $\sqrt{a_3^\perp a_3^\parallel}$ by $O(r^4)$, which is the same order as ϖ_3. So the individual $O(r^4)$ terms of $\lambda_\pm$ are *not* determined by the computation; only their mean $-i\varpi_3$ is. In particular the pair cannot be written as $\pm i\sqrt{84}r^2 - 16ir^4$ to this order: that form omits an unknown term of the same size. What is determined, and what matters, is $a_3^\perp a_3^\parallel = 84r^4 + O(r^6) > 0$. The contour dynamics fits $a_3^\perp = 14r^4(1 + \gamma r^2 + \dots)$ with $\gamma \approx 1.4$ (Table 1, the a^2 coefficient $-115.6 \approx -84\gamma$).

Remark 4.7 (Stability in what sense). Corollary 4.5 is spectral (linear) stability of the steady patch ring, modulo the trivial directions: rotation, the family of rings, and the conserved patch areas. It is not nonlinear stability. Note also that the shape modes carry negative energy (Lemma 3.2), while the centre modes of a stable ring carry positive energy. So the energy–Casimir second variation is indefinite, and Arnold's method does not apply directly. We do not pursue nonlinear stability here.

5 Contour dynamics

5.1 Method

We solve the Euler equations for uniform patches by contour dynamics. Each boundary is a polar graph $\rho_k(\theta)$ about a fixed point c_k on the unit circle, with M equispaced nodes (M odd). The patches are related by the rotation $k \mapsto k + 1$. The velocity is $u = -\frac{q}{2\pi} \oint \log|x - x'|\, dx'$ with vorticity $q = 2/r^2$. The logarithmic self-interaction is integrated with Kress's spectrally accurate weights [10], and derivatives are taken spectrally. The kinematic condition in the frame rotating with the ring gives $\partial_t \rho_k = u_r - u_\theta \rho_k'/\rho_k$.

Steady states. Unknowns are the cosine coefficients of ρ_0 (reflection symmetry) and Ω. The equations are the kinematic condition, the area πr^2, and the position of the vorticity centroid of patch 0 at distance 1; the last fixes the scale of the one-parameter family of rings. We solve by Newton's method, with the Jacobian by complex step [18].

Spectrum. The Jacobian of $\rho \mapsto \partial_t \rho$ is block-circulant. We compute only the columns belonging to patch 0, by complex step, which is exact to round-off. Then we form the Bloch blocks $B_m = \sum_d J_{d,0} \zeta^{-md}$. Two families of trivial directions are factored out exactly: (i) in every block, the patch area is conserved, so $\ell = \rho_0$ (trapezoidal weights) is a left null vector, and we restrict B_m to the invariant subspace $\ker \ell$; (ii) in block $m = 0$, the rotation of the whole ring is a zero mode with a Jordan partner (a change of ring size changes Ω), and we pass to the quotient by this invariant two-dimensional subspace. The residual $|\ell B_m|/|B_m|$ is $\approx 10^{-16}$ and $|B_0 v_{\mathrm{rot}}|/(|B_0||v_{\mathrm{rot}}|) \le 2 \times 10^{-13}$. Code: `code/cd.py`, written for this paper. Its design follows a reviewer's independent code (`code/ngon_reviewer.py`).

Validation. For an isolated circular patch the code returns Kelvin's frequencies $\pm i(n-1)/r^2$ exactly. On every ring the $m = 1$ block contains the translation eigenvalue $i\Omega$ to all printed digits.

The rotation rate satisfies $(\Omega - \Omega_0)/(r/R)^4 \to 1.000$ for $N = 7$ (Theorem 4.3: 1), with an $O(r^2)$ correction. Eigenvalues at $M = 65$ and $M = 129$ (and $M = 193$, 257 near the onset) agree to about 10^{-10}.

5.2 The $(r/R)^4$ coefficients

Numerical result 5.1. For $N = 4, \dots, 8$ and every $1 \le m \le \lfloor N/2 \rfloor$, the slow eigenvalues of the patch ring (contour dynamics, $M = 33$, $r/R = 0.03, \dots, 0.08$) satisfy $(s_m + a^{\perp}_{m,0} a^{\parallel}_{m,0})/\epsilon \to -(a^{\perp}_{m,0} d^{\parallel} + a^{\parallel}_{m,0} d^{\perp})$ and $\varpi_m/\epsilon \to d^{\varpi}$, with the values of Theorem 4.3. The relative errors are at most 8×10^{-5} (Table 1). Here $s = \left(\frac{\lambda_1 - \lambda_2}{2}\right)^2$ and $\varpi = \frac{i}{2}(\lambda_1 + \lambda_2)$ for the two slow eigenvalues $\lambda_{1,2}$ of the block.

[Numerical] `code/cd_asymptotics.py` → `outputs/cd_asymptotics.out`. Fit $c_0 + c_1 a^2 + c_2 a^4$ in $a = r/R$; the table shows c_0.

N	m	c_0 for s (CD)	exact	c_0 for ϖ (CD)	exact
4	1	−3.18751	−51/16	1.00000	1
4	2	−3.75001	−15/4	0.00000	0
5	1	−16.00011	−16	4.00002	4
5	2	−20.00010	−20	3.00002	3
6	1	−56.42447	−8125/144	11.11127	100/9
6	2	−68.61195	−1235/18	12.88906	116/9
6	3	−55.31294	−885/16	0.00000	0
7	1	−156.00547	−156	25.00083	25
7	2	−180.00562	−180	35.00110	35
7	**3**	**−84.00106**	**−84**	**16.00045**	**16**
8	1	−364.46518	−5831/16	49.00360	49
8	2	−397.27904	−1589/4	76.00530	76
8	3	−28.43594	−455/16	55.00345	55
8	4	+189.01734	+189	0.00000	0

Table 1: Contour dynamics against Theorem 4.3. For $N = 8$, $m = 4$ the point block already has $s > 0$ (unstable), and the cores make it more unstable. For $N = 7$, $m = 3$ the point block has $s = 0$, and the cores make $s < 0$ (stable).

5.3 The heptagon at finite core size

Numerical result 5.2. The steady heptagon of uniform patches is found for $0.02 \le r/R \le 0.28019$; continuation in r/R stops there, with the smallest singular value of the Newton matrix falling to zero. For $0.02 \le r/R \le 0.26137$ every Bloch block has spectrum on the imaginary axis to within $|\operatorname{Re}\lambda| \le 2 \times 10^{-10}$ (round-off level), so the ring is linearly stable. At $r/R = a_c \in [0.261377, 0.261379]$ (core radius over side 0.30121; the same at $M = 193$ and $M = 257$) a complex quartet appears in the Bloch blocks $m = 2, 5$, with $\operatorname{Re}\lambda \approx 22.7\sqrt{r/R - a_c}$. A second quartet appears in the blocks $m = 1, 6$ at $r/R \in [0.262194, 0.262197]$ ($M = 129$ and 193; $\operatorname{Re}\lambda = 0.29$ at 0.2624 and 1.07 at 0.265).

[Numerical] `code/cd_heptagon_scan.py` → `outputs/cd_heptagon_scan.out`; `code/cd_onset_modes.py` → `outputs/cd_onset_modes.out`; `code/cd_onset_m1.py` → `outputs/cd_onset_m1.out`. Figures 2, 3.

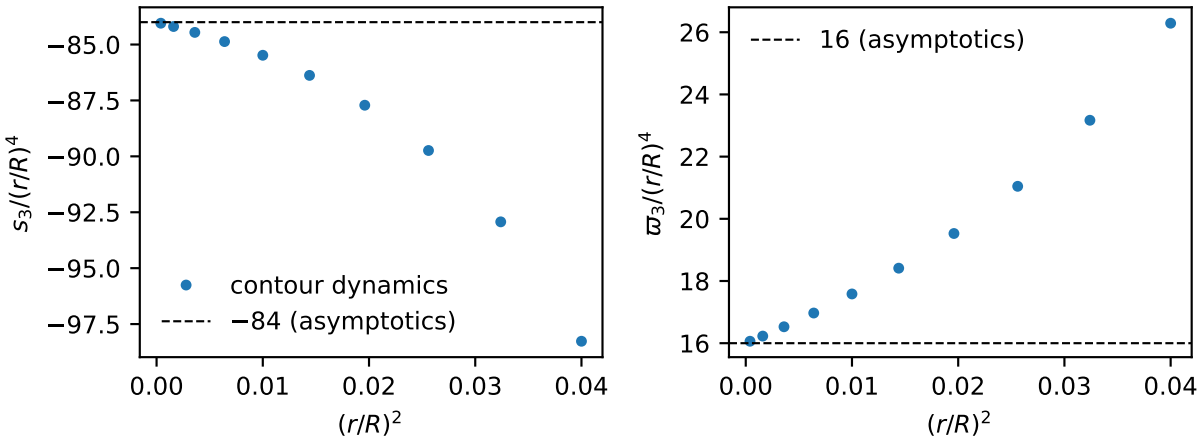


Figure 1: The slow $m = 3$ pair of the patch heptagon from contour dynamics, $r/R \leq 0.2$: $s_3/(r/R)^4 \to -84$ and $\varpi_3/(r/R)^4 \to 16$, as in Corollary 4.5.

Mechanism. In the block $m = 2$ we follow the eigenvalues as r/R increases (Figure 3, right). A displacement mode, whose eigenvector has its boundary perturbation in Fourier modes $k = \pm 1$, moves from $-2.58i$ ($r/R = 0.22$) down to $-4.4i$. An elliptical shape mode ($k = \pm 2$) moves up from $-7.2i$ ($r/R = 0.255$). They meet at about $-4.88i$ at a_c, and leave the imaginary axis as a complex quartet. Just below a_c the two modes are strongly mixed. This is a Krein collision between a centre mode and a shape mode, which have opposite energy signatures (Lemma 3.2). It is unrelated to the order-r^4 mechanism of Corollary 4.5: it happens when the gap between neighbouring boundaries is about one patch radius.

End of the branch. With $M = 257$ we continue the branch to $r/R = 0.28019$. The smallest singular value of the Newton matrix stays at 0.03 up to 0.279 and then drops to 0.009, and Newton's method fails beyond 0.280187. The gap between neighbouring boundaries there is $0.41\,r$. This points to a fold in r/R at about 0.2802; we have not continued past it (`outputs/cd_onset_modes.out`).

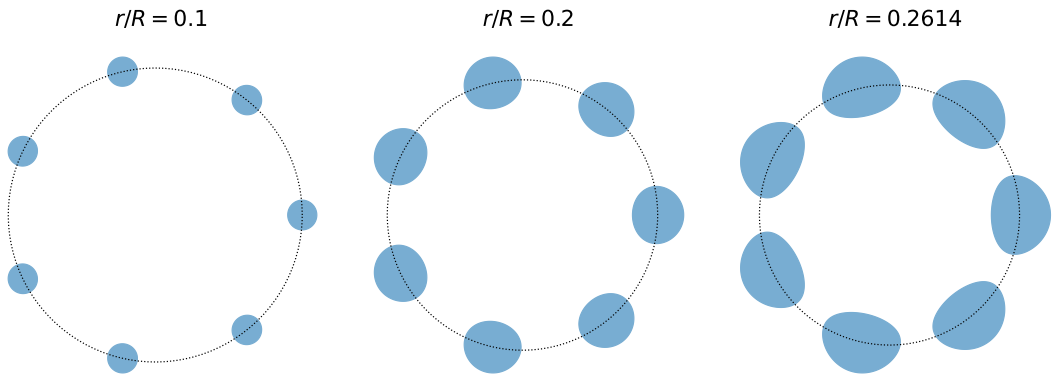


Figure 2: Steady heptagons of uniform patches (contour dynamics) at $r/R = 0.1$, 0.2 and at the onset of instability 0.2614.

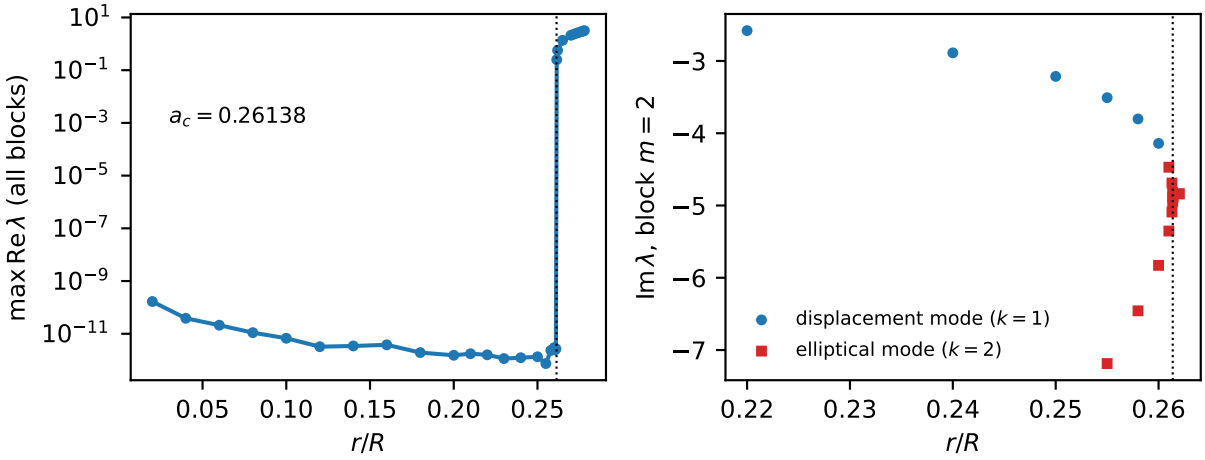


Figure 3: Left: largest real part over all Bloch blocks (after removing the trivial directions) against r/R; it stays at round-off level up to $a_c = 0.26138$. Right: imaginary parts of the eigenvalues of block $m = 2$ in the window $2.5 < |\operatorname{Im}\lambda| < 8$. Circles mark displacement-dominated eigenvectors and squares elliptical ones; they collide at a_c.

5.4 Independent checks by a referee

An independent referee wrote three further checks. We record their outputs here; they were not re-run (`code/ref_cd.py`, `code/ref_lagr.py`, `code/heff_check.py`, `code/fitfreq.py`, unmodified; outputs `outputs/ref_*.out`, `outputs/heff_check.out`).

- *Linear spectrum, second code* [**Numerical**]. It uses the Cauchy form of the velocity kernel, $u - iv = \frac{q}{4\pi}\sum_l \oint_l \frac{\bar{z}'-\bar{z}}{z'-z}\,dz'$, with the trapezoid rule, a fourth-order central finite-difference Jacobian and Bloch blocks, at $M = 65, 97, 129$. Its slow $m = 3$ pair agrees with ours to all printed digits, e.g. $0.3544572i$, $-0.4385848i$ at $r/R = 0.2$, and so do the rotation rates. All blocks are stable up to 0.2613, and the $m = 2, 5$ quartet is present at 0.2615 ($\operatorname{Re}\lambda = 0.25$), so its onset bracket is $(0.2613, 0.2615)$. At 0.265 the blocks $m = 1, 6$ have $\operatorname{Re}\lambda = 1.07$, as in Result 5.2. Its $m = 0$ block shows real parts up to 3×10^{-4} at $M = 97$, because that code does not quotient the rotation Jordan pair (Section 5.5).

- *Nonlinear Lagrangian contour dynamics* [**Numerical**]. Starting from the steady heptagon, the centroids are displaced radially by $10^{-3}\cos(2\pi \cdot 3k/7)$, and the full nonlinear problem is integrated with RK4 ($M = 48$ Lagrangian nodes per patch, $t \le 160$, about nine slow periods). The $m = 3$ displacement stays bounded: its maximum is 5.04×10^{-4} in the first third of the run and 5.01×10^{-4} in the last third at $r/R = 0.20$, and 5.00×10^{-4} and 4.94×10^{-4} at $r/R = 0.15$. Its spectral peaks, at 0.355, -0.439 ($r/R = 0.20$) and 0.202, -0.2225 (0.15), match the linear eigenvalues 0.3545, -0.4386 and 0.2016, -0.2220. The relative patch-area error grows to 2×10^{-4} (0.20) and 10^{-3} (0.15), with a slow drift of the $m = 0$ radial mode, which we attribute to the coarse Lagrangian resolution.

- *The coefficients without lattice sums* [**Numerical**]. Direct 40-digit finite differences of H_{eff} in log-polar coordinates give, for $N = 7$, $\Omega_1 = -1$ and $(d^{\perp}, d^{\parallel}, d^{\varpi}) = (26, 26, 25)$, $(26, 50, 35)$, $(14, 56, 16)$ for $m = 1, 2, 3$, as in Theorem 4.3.

5.5 Spurious real eigenvalues

An earlier independent check with a dense finite-difference Jacobian (reviewer's code, even M) reported, at $r/R = 0.20$–0.22, a single real eigenvalue that grew with resolution; the reviewer read it as round-off in the near-zero cluster. We reran that code at $r/R = 0.20$ ($M = 64$, 96) and analysed the zero cluster with our code. We did not rerun it at 0.21 and 0.22; with our code those points are stable (Result 5.2, with resolution checks at both). Real eigenvalues of this kind come from three sources, all of them trivial directions in exact arithmetic:

1. For even M the Nyquist mode of each boundary is invisible to the spectral derivative. It produces one spurious eigenvalue at 0 per Bloch block. We use odd M.

2. The rotation symmetry gives a 2×2 Jordan block at 0 in the $m = 0$ sector. A perturbation of size δ splits it into $\pm\sqrt{\delta}$. With the finite-difference Jacobian ($\delta \sim 10^{-9}|J|$) this gives a real pair $\pm 4.6 \times 10^{-4}$ (reviewer's units; ± 0.023 in our units) at $M = 64$ and $\pm 3.9 \times 10^{-5}$ at $M = 96$ (`outputs/ngon_reviewer_a020.out`). With the complex-step Jacobian the split is $\leq 10^{-6}$ in our units. After the quotient (ii) above it disappears.

3. The N area modes (one per Bloch block) are neutral only to discretisation accuracy unless area is projected out.

After (i)–(iii), no eigenvalue at $r/R \leq 0.2613$ has $|\operatorname{Re}\lambda| > 2 \times 10^{-10}$, at any resolution tested. The reviewer's other findings agree with ours: the slow pair is imaginary at $r/R = 0.08$ and 0.15, with magnitudes matching $\sqrt{84}(r/R)^2$; $N = 6$ is stable; $N = 8$ is unstable; and there is a strong quartet instability at 0.27.

6 The sphere and the hyperbolic plane

The same construction applies to rings of patches on the unit sphere S^2 and the hyperbolic plane H^2, at height X_0 ($X_0 = \cos$ of the colatitude on S^2, $\cosh$ of the ring radius on H^2). Let $y = X_0^2$. In a stereographic coordinate z with metric $4|dz|^2/(1+K|z|^2)^2$ ($K = \pm 1$), the same adiabatic elimination as in Proposition 3.3, carried out with the conformal factor, gives the effective Hamiltonian (status **Derived**; we state the result without repeating the derivation)

$$H_{\text{eff}} = H + r^4 \sum_k \Big(\frac{1 + K|z_k|^2}{2}\Big)^4 |G_k|^2, \qquad G_k = \frac{1}{2} \sum_{l \neq k} \Big(\frac{1}{z_k - z_l} - \frac{K\bar{z}_k}{1 + K|z_k|^2}\Big)^2,$$

with H the point-vortex Hamiltonian of the Green function G (on H^2, G with its uniform background). We derived the compact form of G_k by hand from $f_{zz} - 2\Phi_z f_z$, where $e^{2\Phi}$ is the conformal factor. For $K = 0$ it reduces to (2). The time-rescaling factor $\cos_K^2(r/2)$ multiplies the symplectic form and the momentum alike, and does not move thresholds.

The same lattice-sum reduction (`code/blocks.py`, now with $\rho = |z|$ symbolic) gives, at the critical mode, a stiffness $a^\perp = \frac{1}{2}[(N-1)(1+y) - \lfloor N^2/4 \rfloor] + \epsilon\, \mathfrak{A}_N(y)/(1-y)^2$ with $\mathfrak{A}_N$ a quadratic polynomial. The stability threshold $y = \Theta_N = \lfloor N^2/4 \rfloor/(N-1) - 1$ moves to $\Theta_N + \mathfrak{D}_N \epsilon$, where $\mathfrak{D}_N = -2\mathfrak{A}_N(\Theta_N)/[(N-1)(1-\Theta_N)^2]$ and the stable side is $y >$ threshold.

Proposition 6.1 (Curved rings, G model)**.** *For $N = 4, \ldots, 12$ (exact rational arithmetic, `code/curved.py`):*

N	*surface*	$\mathfrak{A}_N(y)$	$\mathfrak{D}_N$
4	S^2	$-\frac{15}{16}y^2+\frac{55}{16}y-\frac{9}{8}$	$+\frac{1}{8}$
5	S^2	$-3y^2+\frac{27}{2}y-\frac{11}{2}$	-1
6	S^2	$-\frac{115}{16}y^2+\frac{1547}{36}y-\frac{230}{9}$	$-\frac{380}{9}$
7	–	$-\frac{27}{2}y^2+98y-\frac{141}{2}$ *($\mathfrak{A}_7(1)=14$)*	*no threshold off the plane*
8	H^2	$-\frac{371}{16}y^2+\frac{3319}{16}y-\frac{1519}{8}$	$-\frac{539}{4}$
9	H^2	$-36y^2+\frac{3386}{9}y-402$	$-\frac{244}{3}$
10	H^2	$-\frac{855}{16}y^2+\frac{1315}{2}y-840$	$-\frac{2880}{49}$
11	H^2	$-75y^2+1050y-1525$	-55
12	H^2	$-\frac{1639}{16}y^2+\frac{236255}{144}y-\frac{197725}{72}$	$-\frac{179575}{3528}$

The same polynomial $\mathfrak{A}_N$ serves S^2 and H^2, and $\mathfrak{A}_N(1)$ is the planar value of Corollary 4.4. The general-N forms are (3) *below.*

[Derived (model); Certified (exact algebra given the model)]

General N. Running the same reduction with N and m symbolic, using the closed forms of Lemma 4.1 (valid for $N\ge 4$ and $2\le m\le N-2$, which contains the critical mode), gives on both S^2 and H^2

$$\begin{aligned}
\text{even } N:\quad \mathfrak{A}_N(y) &= -\tfrac{(N-1)(3N^2-12N+10)}{32}y^2+\tfrac{(N-2)(9N^4+179N^3-938N^2+1492N-760)}{2304}y\\
&\quad -\tfrac{(N-2)^2(N-1)(3N^3+6N^2-28N+40)}{2304},\\
\text{odd } N:\quad \mathfrak{A}_N(y) &= -\tfrac{3(N-3)(N-1)^2}{32}y^2+\tfrac{(N-1)^2(9N^3+179N^2-965N+1169)}{2304}y\\
&\quad -\tfrac{(N-3)(N-1)^2(N^3+2N^2-11N+12)}{768},
\end{aligned}\tag{3}$$

$$\mathfrak{D}_N=\frac{(N-2)^2(N-1)(3N^4-104N^3+612N^2-1088N+640)}{288(N^2-8N+8)^2}\qquad (N\text{ even}),$$

$$\mathfrak{D}_N=\frac{(N-3)(N-1)(3N^3-101N^2+509N-539)}{288(N-7)^2}\qquad (N\text{ odd}).$$

Evaluated at $N=4,\dots,12$ they reproduce the per-N exact results of the table above, which `curved.py` computes independently (`outputs/curved_general.out`). **[Certified (exact symbolic computer algebra, `code/curved_general.py`) given the model; not Lean]**. From them: $\mathfrak{D}_N<0$ for $5\le N\le 27$, $N\ne 7$; $\mathfrak{D}_4>0$; $\mathfrak{D}_N>0$ for $N\ge 28$; and the threshold formula, the tabulated values and $\mathfrak{A}_N(1)=$ Corollary 4.4 follow **[Lean: `Heptagon.Curved`, `D_from_A_even`, `D_from_A_odd`, `D_values`, `A_at_one_even`, `A_at_one_odd`, `D_neg_small`, `D4_pos`, `DEven_pos`, `DOdd_pos`]**.

On S^2, finite cores shrink the stable range for $N=4$ ($\mathfrak{D}_4>0$) and enlarge it for $N=5,6$. On H^2 the G model carries a compensating uniform background vorticity. We therefore also treat the kernel $G_1=-\log\tanh(d/2)$, with the same construction (the effective Hamiltonian has the form above with f built from G_1). The point-vortex ring with G_1 has a stability threshold ξ_* in $\xi=\tanh^2(r_{\rm ring}/2)$ for $N\ge 8$, and it moves to $\xi_*+\mathfrak{D}_N^{\tanh}\epsilon$ with $\mathfrak{D}_8^{\tanh},\dots,\mathfrak{D}_{12}^{\tanh}=-26.096064,\ -13.426249,\ -8.0859592,\ -6.6885222,\ -5.3961166$. The thresholds and shifts are computed by symbolic Taylor expansion with the Bloch substitution, as in `blocks.py`, with the sums over the other vortices evaluated directly to 50 digits; the root and its derivative use double-precision secant and central differences (`code/tanh_check.py`) **[Numerical, given the Derived model]**. All five are negative, so the G_1 threshold moves to smaller rings.

7 Relation to prior work

Point vortices. The linear stability of the N-gon is due to Thomson [20], with Havelock's correction [7]: stable for $N \le 6$, neutral for $N = 7$, unstable for $N \ge 8$; see the review of Aref et al. [1]. Nonlinear stability of the heptagon was proved by Cabral and Schmidt [2] and Kurakin and Yudovich [11]. Proposition 2.2 is this classical result.

Small cores, models. An $(r/R)^4$ correction from strain-induced elliptical deformation is standard in the elliptical-moment model of Melander, Zabusky and Styczek [14], and it reflects the small-strain response of Moore and Saffman [15] and Kida [9]. Yang and Kubota [22] found for the counter-rotating pair that the shape changes at order $(r/R)^2$ while the velocity changes only at a higher order; they also report (via a private communication) that Dhanak used an expansion in r/R. Proposition 3.3 is in this tradition, and we do not claim it as new. What we have not found in the literature is the closed form of the order-$(r/R)^4$ blocks for all N and m (Theorem 4.3), the resulting value $a_3^\perp = +14(r/R)^4$ for the heptagon, and a machine check of the algebra.

The heptagon of patches. Here our conclusion contradicts the literature.

- Dhanak [3] (abstract; we could not obtain the full text) treated small uniform cores analytically, with "approximate equations for the shape and motion of a vortex subjected to an external velocity field". He found "the system is stable if $N < 7$ and unstable if $N \ge 7$", in general agreement with Dritschel [4] "except in a region of parameter space where a high degree of accuracy is required in the numerical calculation to resolve the growth rate of small disturbances".

- Polvani and Dritschel [16, p. 52]: "when $N = 7$, any finite size leads to instability – see Dhanak 1992".

- Llewellyn Smith and Crowdy [12]: in Dritschel's study [4] the heptagon "becomes unstable to a displacement instability when the point vortices are replaced by vortex patches, even very small ones ... Dhanak (1992) argues that there are, in fact, two such instabilities." We could not read Dritschel's paper either.

Contrary to the statement in Dhanak (1992), repeated in Polvani and Dritschel (1993), we find that small uniform cores make the heptagon linearly stable. The evidence has two independent parts: the order-$(r/R)^4$ blocks (Theorem 4.3, given Proposition 3.3), and the contour dynamics. The latter reproduces the coefficient -84 to 10^{-5} and finds the spectrum on the imaginary axis to round-off up to $r/R = 0.2614$. An independent referee's code, a nonlinear simulation (Section 5.4), and a reviewer's code agree. Since we have not read Dhanak's paper we cannot locate the discrepancy. A reader with access might check: (i) the order of the expansion: whether the decisive term for $N = 7$ is computed at order $(r/R)^4$, and whether an $(r/R)^2$ term appears, which would contradict Lemma 3.1; (ii) the treatment of the shape. For $N = 7$, $m = 3$ the pieces of Proposition 4.2 are $\mu = -1$, $\Lambda_+ = 10$, $\Lambda_- = 6$, $\kappa = 22$, so

$$d^\perp = \tfrac{1}{4}(\Lambda_+^2 + \Lambda_-^2) + \mu\kappa + 2\mu^2 = 34 - 22 + 2 = 14, \qquad d^\parallel = 34 + 22 = 56.$$

The positive part 34 comes from the term $|L_k|^2$, i.e. from the ellipses *following* the perturbed strain. A model that freezes each ellipse at its equilibrium shape (shape amplitude fixed at $r^2\overline{M_k^0}$) keeps the same rotation rate but loses exactly this term: it gives $d^\perp = -20$, $d^\parallel = 22$, so $a_3^\perp a_3^\parallel < 0$ and the heptagon would be *unstable*. Dropping the $|L|^2$ term for any other reason has the same effect. We verified both numbers by direct 40-digit differentiation of the two Hamiltonians (`code/frozen_check.py` → `outputs/frozen_check.out`). This is a concrete candidate for where

an approximate shape model could differ from ours. Reversing the sign of the whole r^4 term would also give -14. The contour dynamics confirms the slaved value (Table 1); (iii) the definition of stability: whether "unstable" refers to the mode $m = 3$ or to a resonance with shape modes at finite r, and whether area, rotation and Jordan directions are factored out (Section 5.5 shows how these produce spurious real eigenvalues); (iv) whether the finite-r instability of Result 5.2 (onset at core radius over side 0.301) is what is meant. The "two such instabilities" of Dhanak, as reported, may be a displacement–shape resonance of this kind. Finally, Dhanak's abstract places his disagreement with Dritschel only "in a region of parameter space where a high degree of accuracy is required in the numerical calculation to resolve the growth rate of small disturbances". This is consistent with the scales here: the slow $m = 3$ frequencies are tiny, $\sqrt{84}(r/R)^2 \approx 0.023$ at $r/R = 0.05$ (our units, against an orbital frequency 3), so a weak spurious growth or decay of that size is easily produced by an under-resolved computation. Wan [21] studied the stability of rotating vortex patches; we have not read it.

Existence. García [5, Thm. 1.1] proves existence of rotating patch polygons near Thomson polygons for every $N \geq 2$, including $N = 7$ (Remark 3.4); Hmidi–Mateu [8] and Hassainia–Wheeler [6] treat related multipole patch equilibria. None of these addresses linear stability.

8 Status and limitations

- **The effective Hamiltonian is formal.** Proposition 3.3 (adiabatic elimination of the fast elliptical modes, remainders order-counted) is not a theorem. Consequently the stability of the small-core heptagon (Corollary 4.5) is *Derived*, not proved. A proof would need a rigorous reduction of the patch dynamics near the steady ring to a slow manifold, with error bounds of order r^6 in the block.

- **Existence** of the steady patch heptagon for small patches is García's theorem, which covers $N = 7$ (Remark 3.4); it gives boundaries at least C^1, and the smoothness used in our expansions is not checked beyond that. Numerically the branch exists up to $r/R \approx 0.2802$.

- **Linear only.** We say nothing about nonlinear stability; the energy–Casimir method fails here.

- **What Lean checks:** the lattice sums for all N, the block polynomials and the critical-mode signs for all N, the heptagon values, the spectral criterion of a block, the algebra of the elimination, and Newton's theorem for the logarithm on circles. **What Lean does not check:** the Taylor expansion and Bloch evaluation that turn H_{eff} into lattice sums (Proposition 4.2; by hand, and cross-checked by computer algebra), the shape energy (Lemma 3.2; by hand), the adiabatic elimination itself, and all numerics.

- **Numerics** are floating point (spectral contour dynamics). They were checked for resolution ($M = 33$–257), against exact solutions, and against the asymptotics, but they carry no error bounds.

- **The curved-surface coefficients** (Section 6) depend on the curved effective Hamiltonian, which we state without a full derivation (only the compact form of G_k is derived here). They were not checked by contour dynamics here.

- **Prior work** could not be read in full (Dhanak 1992, Dritschel 1985, Wan 1986). The conflict is reported as found.

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A Proof of Proposition 4.2

Write $u_k = s_k + it_k$, so $z_k = \zeta^k e^{u_k}$, and $c_j = 1-\zeta^j = 1/p_j$. Every term of F is a sum over k of a local term; by the shift symmetry, B_m is obtained from the local terms at $k=0$. Write ∇^2_{ab} for the second derivatives with respect to (s_a,t_a) and (s_b,t_b).

The momentum. $J = \frac12\sum e^{2s_k}$ contributes $\partial_{s_0}J = 1$, and $-\Omega\left(\begin{smallmatrix}2&0\\0&0\end{smallmatrix}\right)$ to every B_m.

The point-vortex energy. For the pair $(0,j)$, $\log(z_0-z_j) = \log c_j + \log(1+p_j\delta)$ with $\delta = (e^{u_0}-1) - \zeta^j(e^{u_j}-1)$. The quadratic part of $\log(1+p_j\delta)$ is $\frac12 p_j(u_0^2-\zeta^j u_j^2) - \frac12 p_j^2(u_0-\zeta^j u_j)^2 = -\frac12\zeta^j p_j^2(u_0-u_j)^2$. Here we used $1-p_j = -\zeta^j p_j$ and $1+\zeta^j p_j = p_j$ (the coefficients of u_0^2, u_j^2, u_0u_j are $-\frac12\zeta^j p_j^2$, $-\frac12\zeta^j p_j^2$, $\zeta^j p_j^2$). Since $g_j := \zeta^j p_j^2 = -1/(4\sin^2(\pi j/N))$ is real, the quadratic part of $H = -\sum_{k<l}\operatorname{Re}\log(z_k - z_l)$ is $\frac12\sum_{k<l} g_{l-k}\big[(s_k-s_l)^2 - (t_k-t_l)^2\big]$. Hence $B_m(H) = \operatorname{diag}\big(g(0)-g(m),\, -(g(0)-g(m))\big)$ with $g(n) = \sum_j g_j\zeta^{jn} = T_2(n) - T_1(n)$. The linear part gives $\partial_{s_0}H = -\operatorname{Re}\sum_j p_j = -\frac{N-1}{2}$, so $\Omega_0 = -\frac{N-1}{2}$ and $-2\Omega_0 = N-1$.

The core term. Let $m_k = \zeta^{2k}M_k = \sum_{j\neq 0}(e^{u_k} - \zeta^j e^{u_{k+j}})^{-2}$, so $|m_k| = |M_k|$. Expanding $(c_j+\delta)^{-2} = p_j^2(1 - 2p_j\delta + 3p_j^2\delta^2 - \dots)$ gives $m_k = \mu + L_k + q_k + O(u^3)$, with $\mu = T_2(0)$ real,

$$L_k = \sum_l \lambda_{l-k}u_l, \quad \lambda_0 = -2\sum_j p_j^3,\ \lambda_j = 2\zeta^j p_j^3\ (j\neq 0), \qquad q_k = \sum_j \big[-p_j^3(u_k^2 - \zeta^j u_{k+j}^2) + 3p_j^4(u_k - \zeta^j u_{k+j})^2\big].$$

Then $|m_k|^2 = \mu^2 + 2\mu\operatorname{Re}L_k + |L_k|^2 + 2\mu\operatorname{Re}q_k + O(u^3)$. Set $\Lambda(n) = \sum_j \lambda_j\zeta^{nj} = -2(T_3(0) - T_3(n+1))$. It is real, because $\lambda_{-j} = \bar\lambda_j$. For $|L|^2 = \sum_k L_k\bar L_k$ we have $\partial_{s_a}\partial_{s_b} = \partial_{t_a}\partial_{t_b} = 2\operatorname{Re}\sum_k \lambda_{a-k}\bar\lambda_{b-k}$ and $\partial_{s_a}\partial_{t_b} = 2\operatorname{Im}\sum_k \lambda_{a-k}\bar\lambda_{b-k}$. Summing against ζ^{mb} gives the block contributions $\Lambda(m)^2 + \Lambda(-m)^2$ on the diagonal and $i(\Lambda(m)^2 - \Lambda(-m)^2)$ off the diagonal. For $\operatorname{Re}q = \operatorname{Re}\sum_{a,b}K_{ab}u_au_b$, with K real and symmetric by the reflection symmetry, the contributions are $\pm 2\kappa$ on the diagonal ($+$ for ss, $-$ for tt) and 0 off it. Here $\kappa = \sum_b K_{0b}\zeta^{mb}$ is q_0 evaluated bilinearly on $(u_a) = (\zeta^{-ma})$, $(u_b) = (\zeta^{mb})$:

$$\kappa = \sum_j \big[-p_j^3(1-\zeta^j) + 3p_j^4(1-\zeta^{j(1-m)})(1-\zeta^{j(1+m)})\big] = -T_2(0) + 3\big(T_4(0) - T_4(1-m) - T_4(1+m) + T_4(2)\big).$$

Multiplying by $\frac14$ (from $\Phi = \frac14\sum|m_k|^2$) and by 2μ where appropriate gives the bracketed terms. Finally, $\partial_{s_0}\Phi = \frac12\mu\Lambda(0) = -\mu T_2(0) = -\mu^2$ (by Lemma 4.1(b)), so $\Omega_1 = -\mu^2$ and $-2\Omega_1 = 2\mu^2$. □

B Numerical details

Steady states are converged to residual $\le 2\times10^{-12}$; the smallest singular value of the Newton matrix stays ≥ 0.03 for $r/R \ge 0.2$. The resolution schedule is $M = 65$ for $r/R \le 0.205$, 129 up to 0.255, and 193 beyond, with checks at 129, 193 and 257; the branch continuation beyond 0.278 uses $M = 257$ with the step refined down to 10^{-5}. The onset was bisected to width 2×10^{-6} at $M = 193$ and $M = 257$ with the criterion $\max\operatorname{Re}\lambda > 10^{-7}$. Run times are minutes per point on a workstation; `run_all.sh` with the option `--fast` reproduces a reduced set.

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